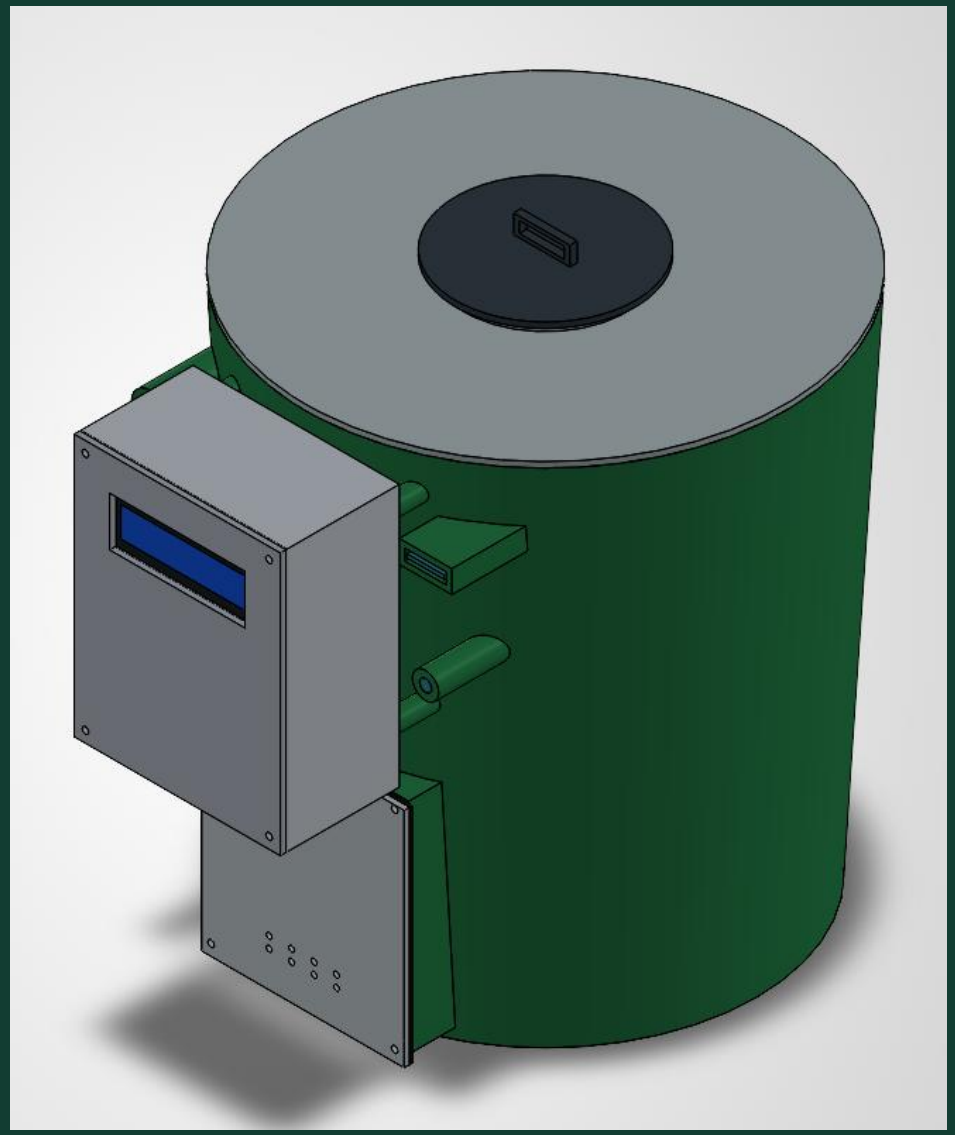
SMART COMPOSTING SYSTEM

Arduino-based monitoring and motor-assisted mixing

10 L BUCKET • SENSORS • DC MIXER



Meet the project team

Four members working on an Arduino-based smart composting project.

01 Mayur Santosh Darekar

PRN 2126UMEM1024

02 Sairaj Hon

PRN 2126UMEM1012

03 Abhinav Bothe

PRN Not provided

04 Pratik Bankar

PRN 2126UMEM1022

Why smart composting?

Help users monitor a small organic-waste batch and reduce manual mixing effort.

01

The problem

Unmonitored moisture and limited airflow can make a small compost batch difficult to manage.

02

The objective

Combine temperature and moisture sensing with Arduino-based control and powered mixing.

03

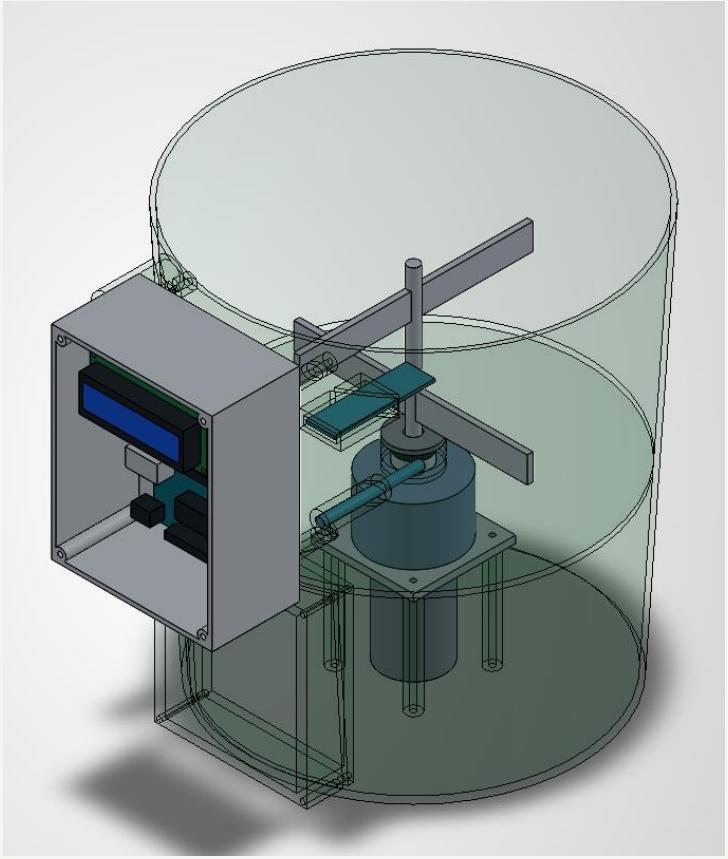
The scope

A compact 10 L bucket prototype for household-scale learning and demonstration.

Composting still depends on suitable feedstock, air, moisture and time—not electronics alone.

The system at a glance

A composting chamber, a lower drive area and an external electronics enclosure.



Transparent CAD view showing the proposed internal arrangement

Sense

DHT11 and soil moisture sensor provide local condition readings.

Control

Arduino Uno reads inputs and sends commands to the L298N driver.

Mix

A 775 DC motor drives the proposed shaft-and-paddle assembly.

Hardware used in the project

The following components were specified by the project team.

Arduino Uno
Central controller for sensor input and motor commands

12 V / 2 A adapter
Specified DC power supply; load suitability unverified

DHT11 sensor
Local air temperature and relative humidity sensing

Voltage regulator
Regulated electronics supply; model/output unspecified

Soil moisture sensor
Relative wetness sensing; calibration required

10 L bucket
Nominal container volume for the composting prototype

L298N motor driver
Interface between Arduino control signals and motor

Partition plate
Separates the upper chamber and lower drive space

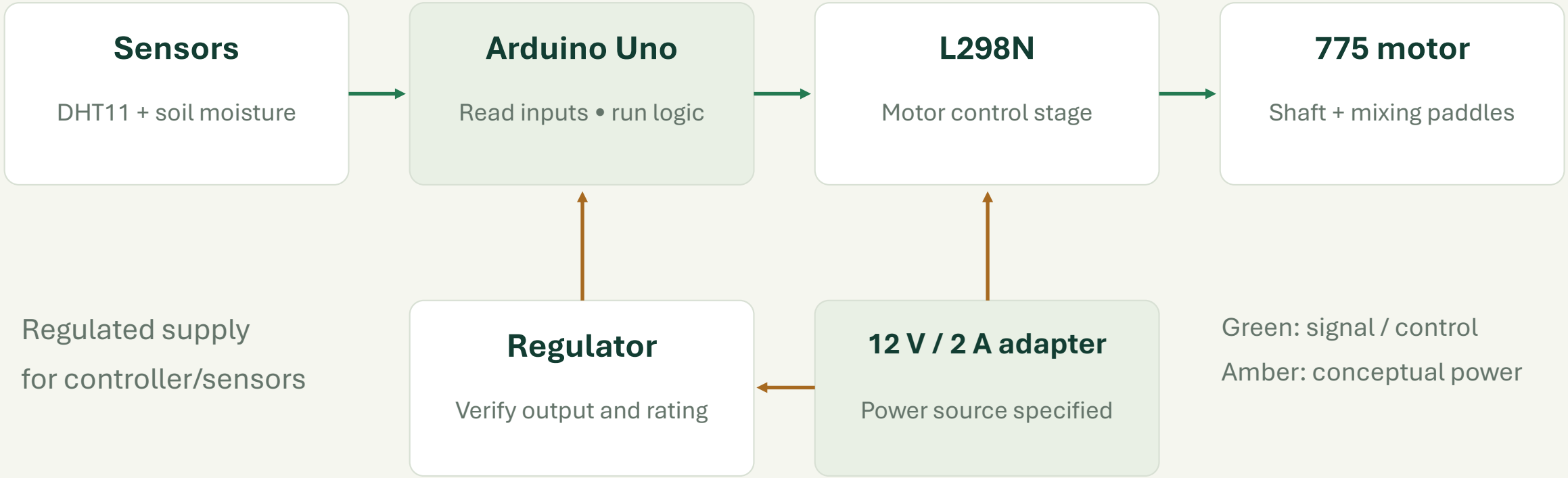
775 DC motor
Proposed source of rotation for the mixing mechanism

Jumper wires
Low-current signal and prototype interconnections

The shaft, paddles, enclosure and display are shown in CAD; exact build details were not supplied.

Functional block diagram

Conceptual signal and power paths—not a pin-level wiring diagram.



Common signal ground is required. Never power the motor from an Arduino I/O pin.

Proposed working sequence

An explanatory operating cycle; actual firmware timing and thresholds were not supplied.

01

Load

Add suitable organic waste and dry carbon-rich material.

02

Sense

Read temperature and relative moisture conditions.

03

Evaluate

Apply programmed timing and check sensor validity.

04

Mix

Run a controlled mixing cycle with the lid secured.

05

Review

Log observations and manually adjust moisture or feedstock.

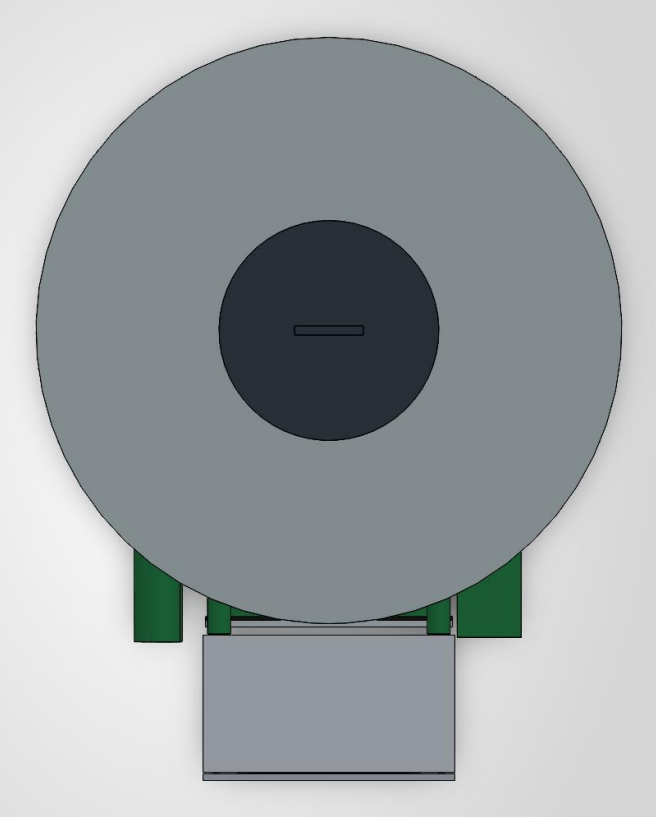
No pump, heater, fan or wireless module was specified; those functions are not claimed as implemented.

Complete assembly: front and top

Conceptual CAD views—not to verified 10 L scale. Vessel, lid, electronics box and lower access cover.



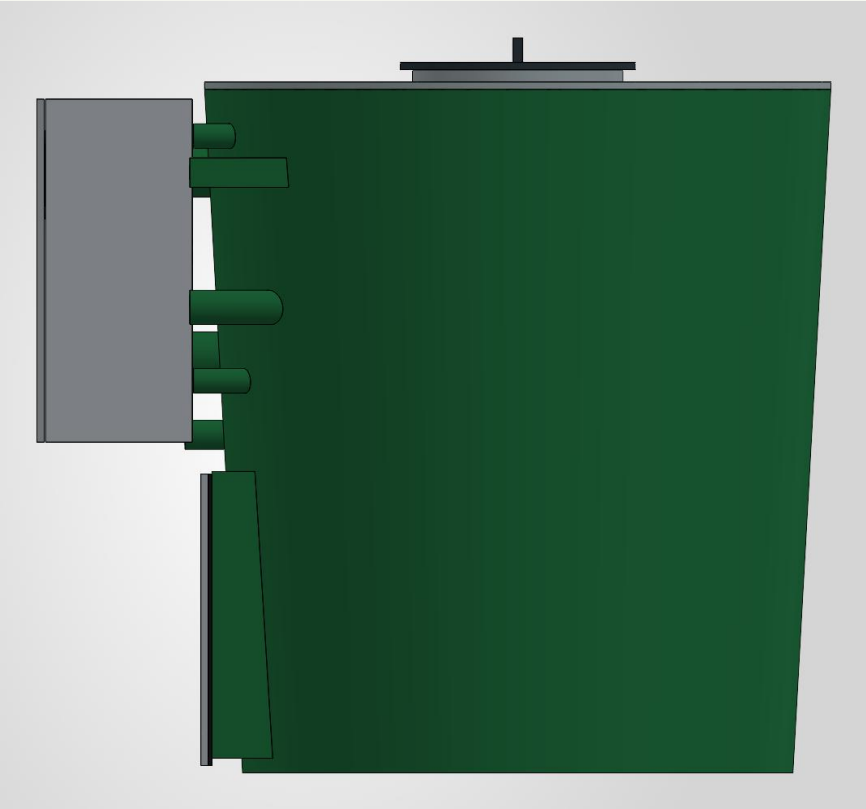
Front view • control enclosure and lower access cover



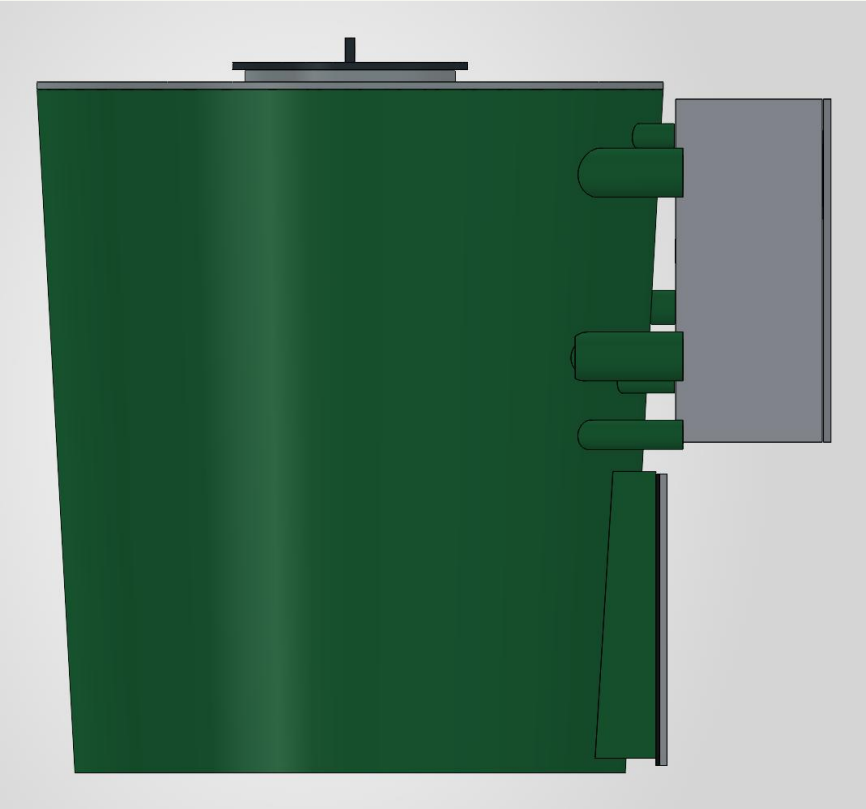
Top view • main lid and central access hatch

External side views

Both sides show enclosure projection, side penetrations and maintenance access.



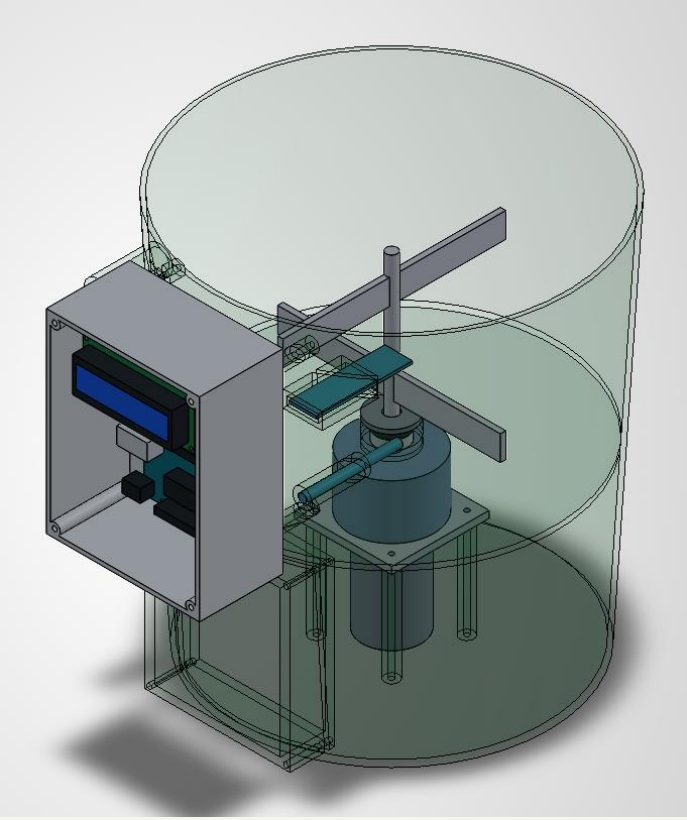
Right-side CAD view



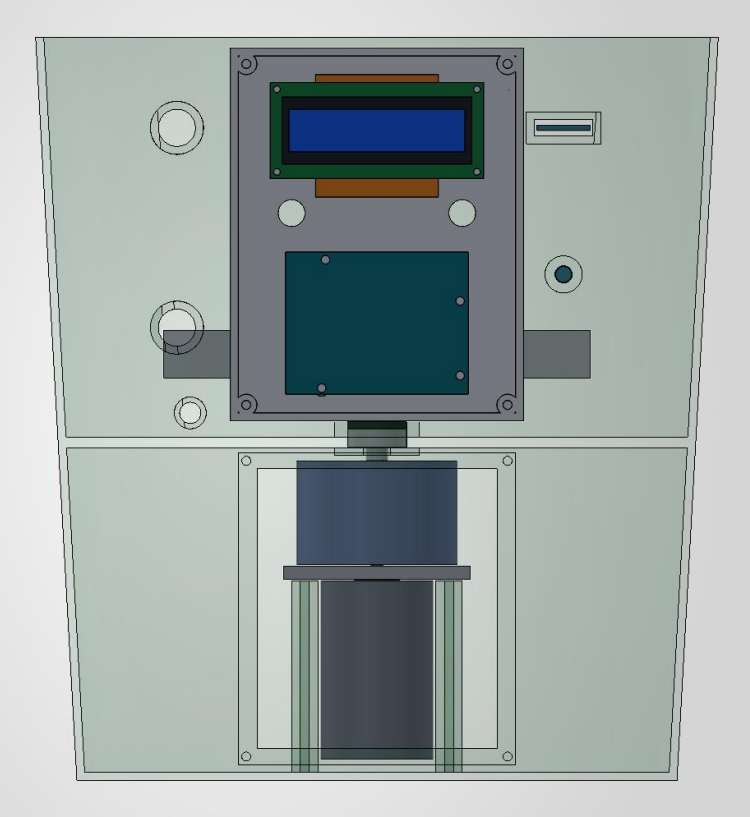
Left-side CAD view

Internal arrangement

Transparent views expose the relationship between the mixer, partition and drive area.



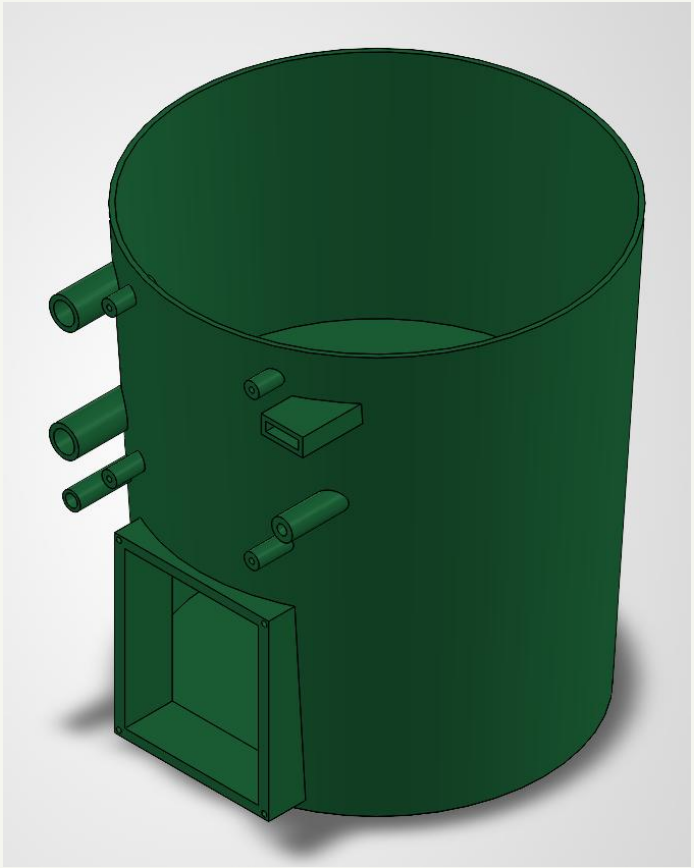
Isometric view • mixer and electronics layout



Front view • upper chamber and lower motor region

Bucket body and partition

Nominal bucket size: 10 L, as specified by the project team.

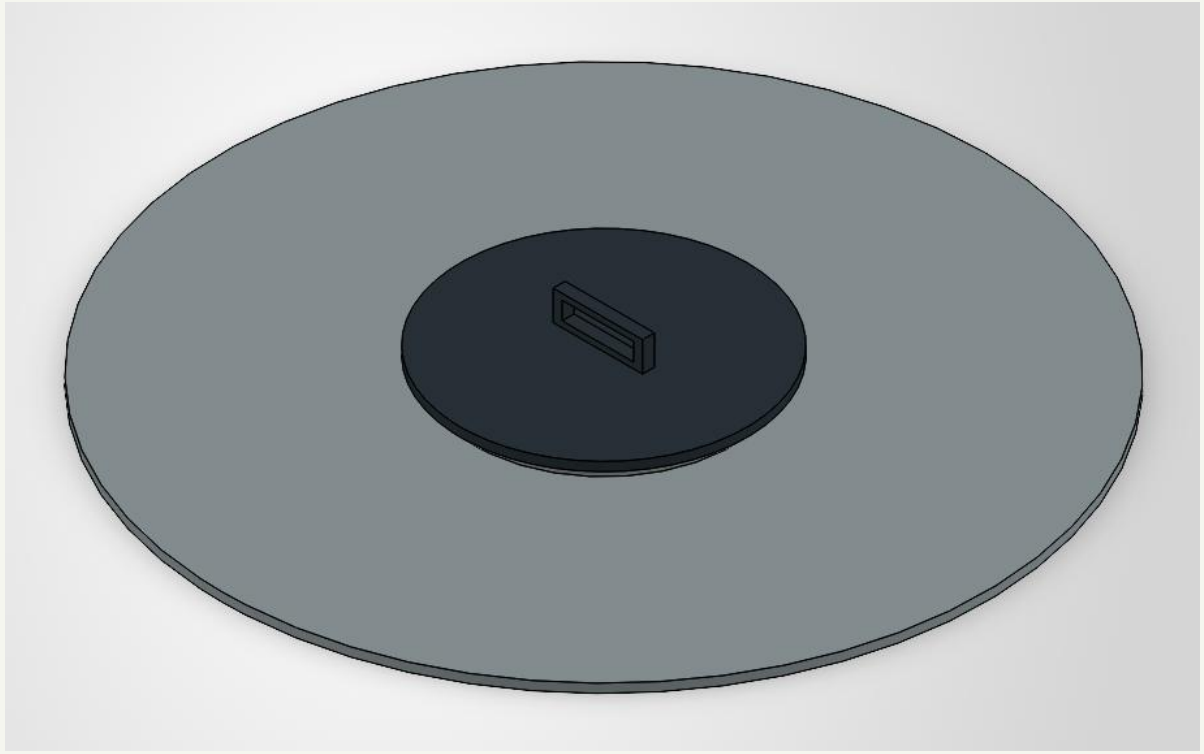


Bucket body with the internal partition shown in CAD

- Upper space holds the organic material being processed.
- The partition separates the compost chamber from the proposed drive region.
- Side features accommodate the illustrated interfaces and enclosure mounting.
- Usable compost volume is less than nominal bucket capacity.

Removable lid and access hatch

The top closure supports loading, inspection and cleaning of the chamber.



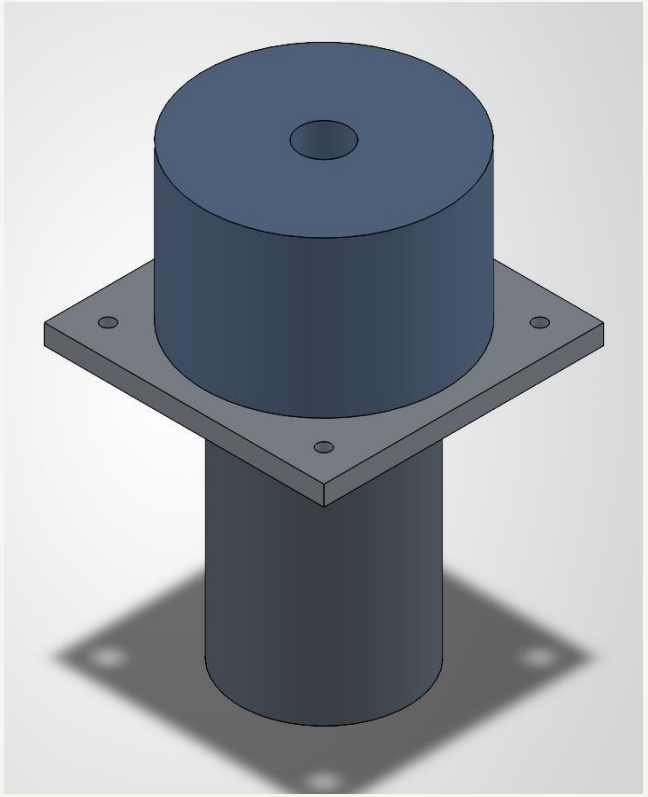
CAD detail • lid and central handled hatch

- Removable main lid provides access to the vessel.
- A handled central hatch is visible in the supplied design.
- Keep hands clear and disconnect power before opening or servicing.

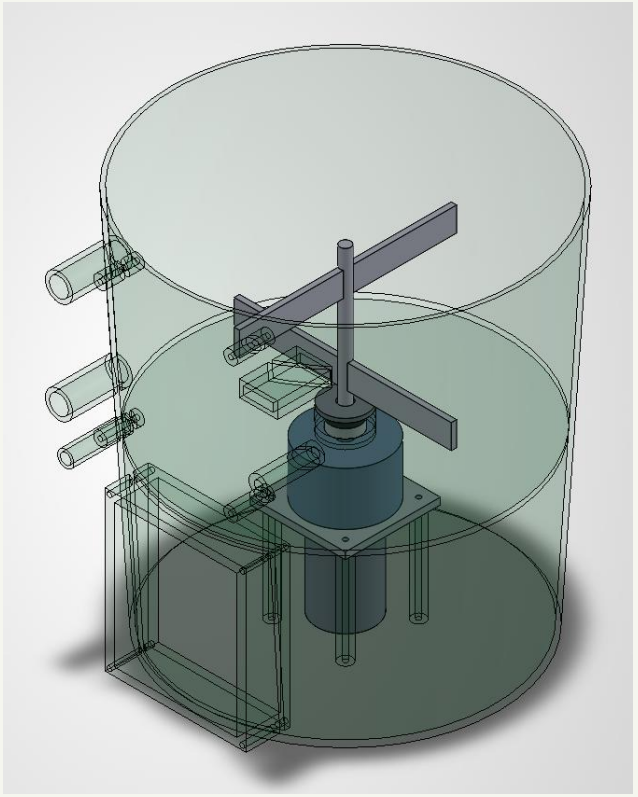
A lid interlock is a recommended improvement; it was not listed among the installed components.

Motor, mounting and drive space

The 775 DC motor is intended to rotate the internal mixing assembly.



Motor/mount CAD detail • simplified envelope geometry

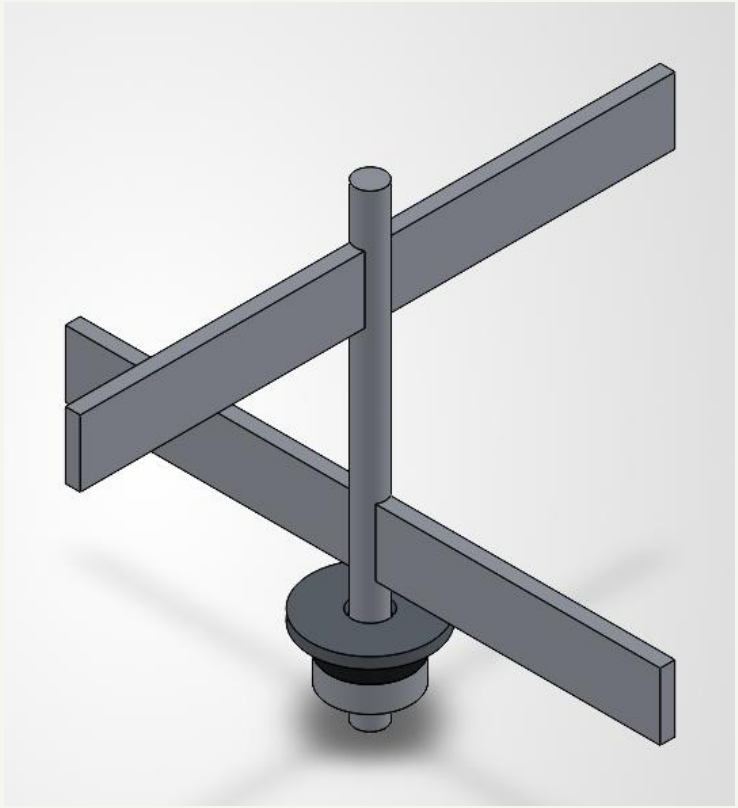


Drive area positioned below the partition

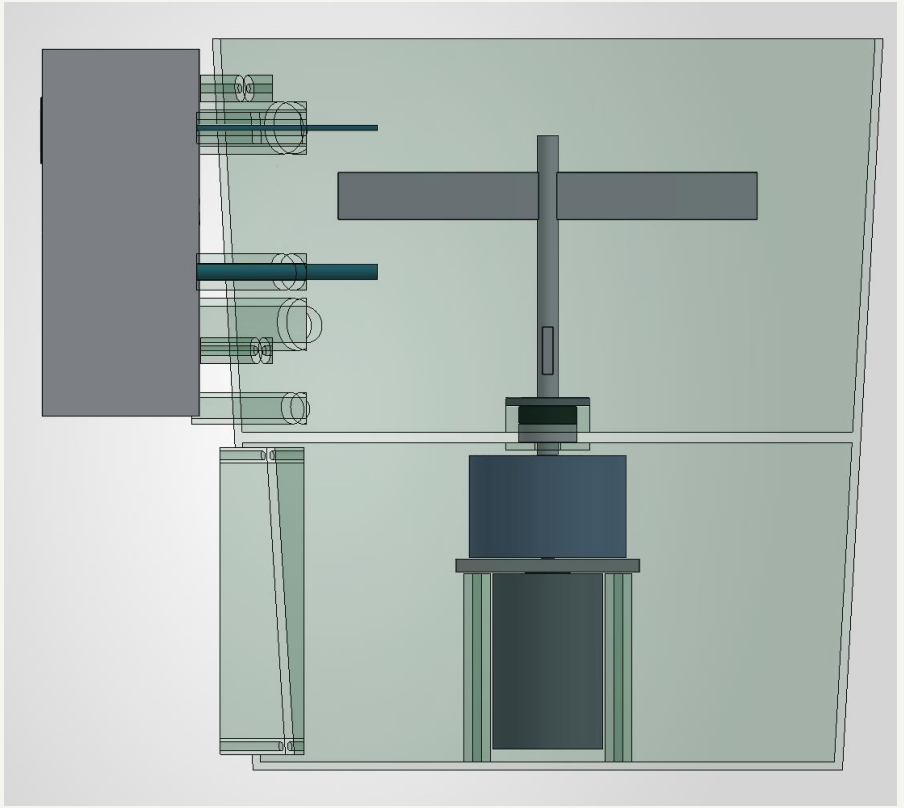
Required speed, torque, reduction ratio and start/stall current remain to be verified.

Mixer shaft and paddles

The illustrated mechanism distributes mixing action across two heights.



Shaft-and-paddle assembly with a lower coupling/seal detail

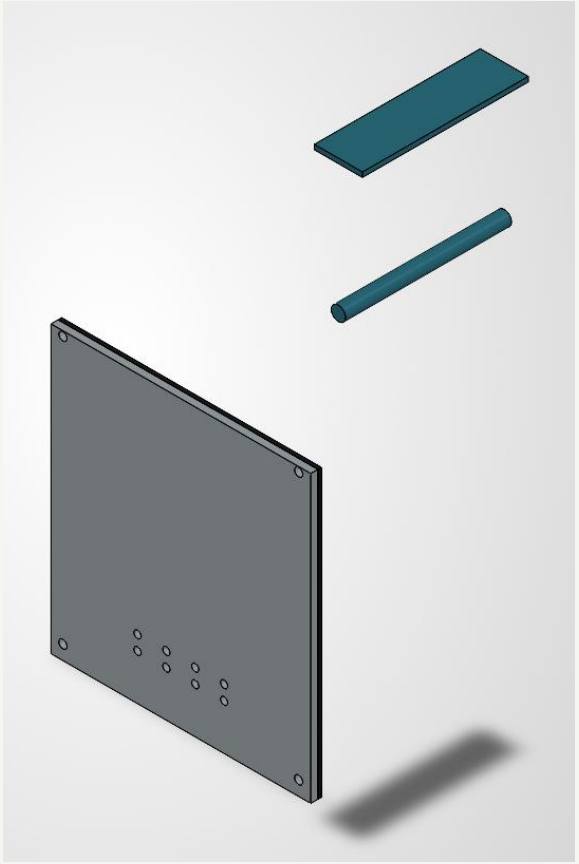


Transparent side view: vertical drive alignment

Check paddle clearance, shaft support, attachment strength and leakage before loaded operation.

Sensors and service access

Sensor-like geometry and the removable lower cover are shown as simplified CAD features.

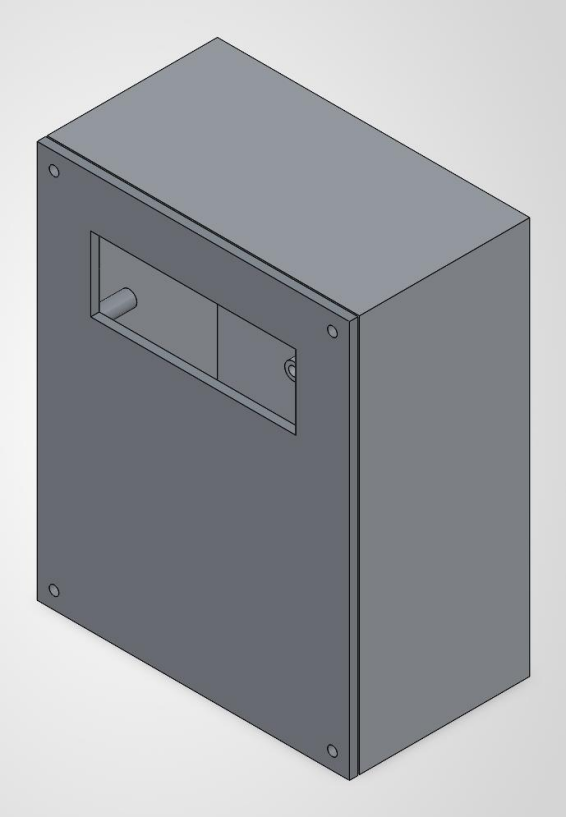


Access cover and simplified sensor representations

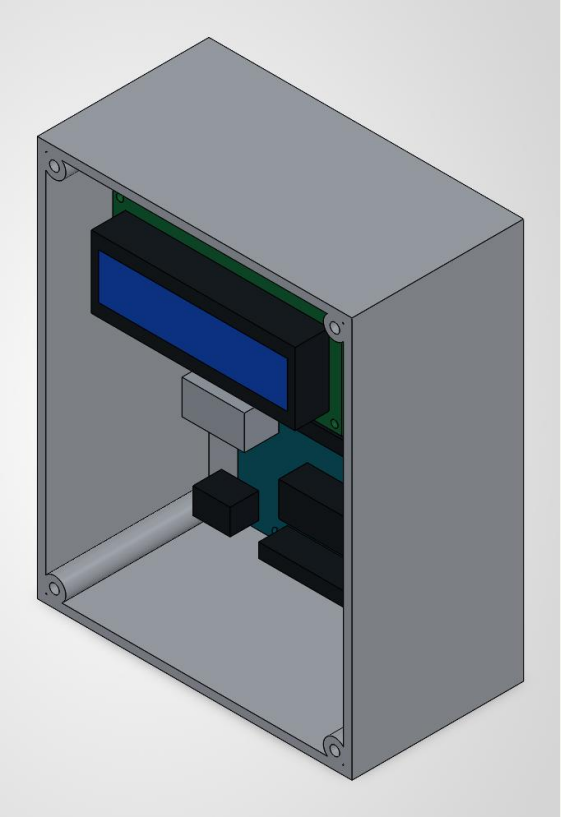
- DHT11: measures local air temperature and relative humidity.
- Soil moisture sensor: indicates relative wetness after calibration.
- Protect electronics from wet waste, condensation and moving paddles.
- Remove power before opening the lower service cover.

Electronics enclosure

The enclosure organizes the control electronics outside the composting chamber.



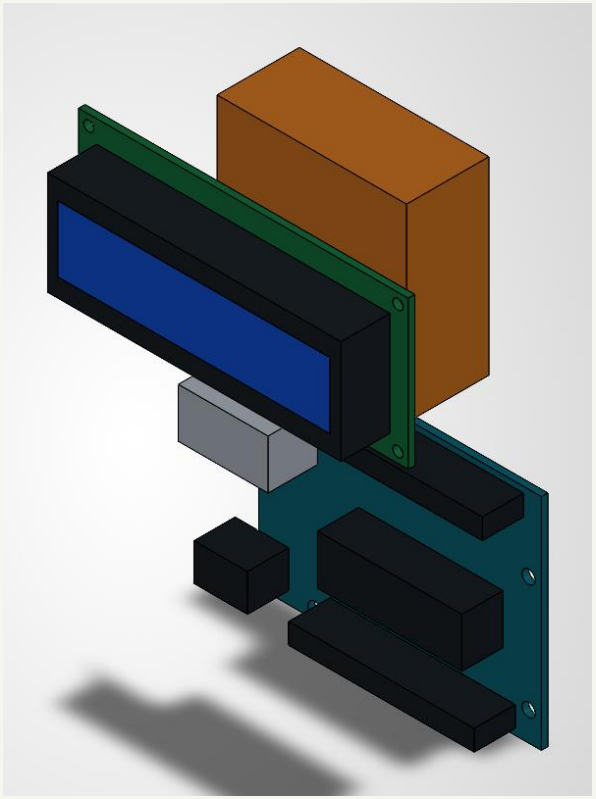
Closed enclosure • front-panel display opening



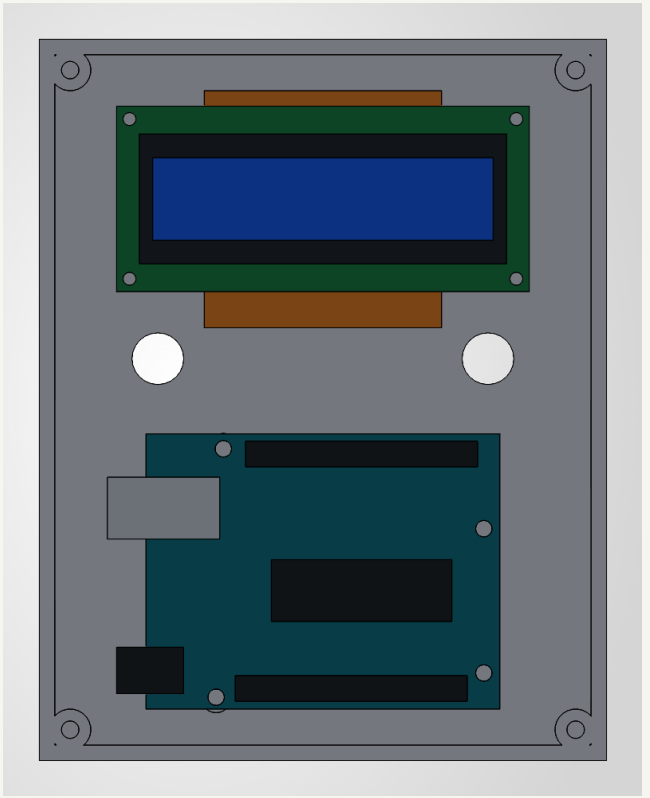
Open enclosure • illustrated electronics arrangement

Controller and display layout

Simplified CAD boards show placement rather than a verified circuit or PCB design.



Board/display subassembly • simplified CAD geometry



Front view of the electronics inside the enclosure

LCD display: visible in CAD, but not confirmed in the team’s supplied hardware list.

What the sensors can—and cannot—tell us

Measurements need calibration, correct placement and interpretation.

01

DHT11

Typical range: 0–50 °C and 20–80% RH.

Keep dry and protected. It is not a waterproof compost-core probe.

02

Soil moisture

Relative wetness only.

Calibrate with the actual material; raw readings are not exact water content.

03

Arduino control

Can read sensors and command mixing.

No Wi-Fi, automatic watering or automatic heating hardware was specified.

A small 10 L unit is not guaranteed to reach thermophilic conditions or sanitize compost.

Power compatibility and safety

The listed motor, driver and adapter are not yet a verified compatible power system.

Motor current

Check the exact 775 motor’s starting, running and stall current against the supply and driver.

L298N heating

Allow for driver voltage drop and heat. Module current capability depends on cooling and duty.

Regulator and wiring

Verify regulated output, polarity and heat. Use load-rated motor wiring and appropriate protection.

Wet and moving parts

Guard the mixer, isolate wet areas and stop power before access. Confirm shaft sealing and support.

Do not assume a 12 V / 2 A adapter or common L298N module can safely start every 775 motor.

Validation plan

No measured performance results were supplied. The following checks are proposed.

CHECK	METHOD	RECORD
Temperature	Compare with a suitable reference thermometer.	Error and repeatability
Moisture response	Compare repeatable dry, damp and wet samples.	Raw readings and calibration
Mixer operation	Use guarded empty and progressively loaded runs.	Mixing, clearance and jams
Electrical loading	Measure safe start/run current and supply voltage.	Voltage sag and thermal rise
Partition / enclosure	Check for leakage, splash entry and loose fittings.	Observed issues and fixes
Composting trial	Log batch inputs, conditions and maturation over time.	Trends—not assumed results

Only report composting time, energy use, reliability or compost quality after actual measurement.

Expected benefits and applications

Potential advantages to evaluate through prototype testing—not established performance claims.

More visible conditions

Sensor readings can help the user identify changing temperature and moisture conditions.

Less manual mixing

Motor-assisted stirring is intended to reduce the effort needed for batch mixing.

Compact demonstration

A 10 L bucket format suits student laboratories and household-scale learning.

Integrated engineering

The project connects mechanical construction, embedded control and organic-waste management.

Future improvements

Upgrade only after the basic sensing, mixing and electrical system are validated.

01	**Reliable sensing**	Use a suitable protected temperature probe and calibrate the moisture measurement.
02	**Matched drive system**	Select motor reduction, driver and power supply from measured torque/current needs.
03	**Safer access**	Add lid interlocking, overload protection and improved sealing around the shaft.
04	**Data and automation**	Add local logging or wireless monitoring; consider watering only with suitable hardware.

References and project evidence

Primary documentation supports the technical descriptions and stated limitations.

Arduino UNO R3 documentation
https://docs.arduino.cc/hardware/uno-rev3/
Controller overview and 5 V logic

STMicroelectronics — L298 datasheet
https://www.st.com/resource/en/datasheet/l298.pdf
Driver ratings, voltage drop and thermal considerations

Adafruit — DHT sensor guide
https://learn.adafruit.com/dht/overview
DHT11 measurement ranges and sensor characteristics

US EPA — Composting At Home
https://www.epa.gov/recycle/composting-home
Feedstock, moisture and aeration fundamentals

Project evidence: team details, stated hardware and all 18 supplied CAD images. No test data supplied.

A practical step toward smarter composting

Monitor conditions. Assist mixing. Validate the system.

Small prototype.

Clear engineering goals.

- Arduino Uno integrates sensing and motor commands.
- The 10 L bucket concept combines a partition and mixing assembly.
- Next step: verify safe operation and measure composting performance.

THANK YOU / QUESTIONS?

